

■ Chart Generator — Final Spec (Energy Nation Data Briefs)

1) Goals

- Produce **one or two crisp visuals per brief**.
- Ensure **deterministic & reproducible outputs**.
- Output **lightweight PNGs** sized for Substack/Twitter/LinkedIn.

2) Inputs

- **Data:** `energy_nation_flat_dataset_briefs_ready.csv` (single source of truth).
- **Query parameters:**
 - `topic` (free text → slugified for filenames: lowercase, underscores, strip special chars).
 - `filter`: dict (e.g., `{"province":["AB","ON"],"sector":["Energy"]}`).
 - `scenario`: `base | mpo | indigenous | combined | override_{0.25|0.5|0.75|1.0}`.
 - `compare`: **exactly two Unique ID's** for side-by-side/dumbbell plots.
 - `metric`: one of `EV`, `delta_EV`, `blended_prob`, `delta_p`, `crosser`, `dropout`.
 - `agg`: `"sum" | "mean" | "median" | "count"`.
- **Column map (canonical):**
 - Base: `EV_disc_base` (preferred) else `EV_base`; `blended_prob`
 - Scenarios: `EV_mpo`, `EV_indigenous`, `EV_combined`, `EV_disc_override_*`
 - Deltas: `delta_EV_*`, `delta_p_*`
 - Flags: `crosser_*`, `dropout_*`
 - Meta: `Unique ID`, `project`, `company`, `province`, `sector`, `group`, `project_cost`

3) Supported Chart Types

1. **Portfolio EV Uplift — Bar (Top-N)**
2. **Distribution — Histogram of ΔEV**
3. **Cohort Crossing — Count Bars**
4. **Project Side-by-Side — Dumbbell Plot**
5. **Province/Sector Aggregates — Grouped Bar**
6. **Δp vs Project Cost — Scatter**

4) Output & File Rules

- Size: 1100 px wide, 220–700 px height
- Format: PNG (<200 KB), optional WebP duplicate
- Naming: `charts/YYYYMMDD\_slugifiedtopic\_scenario\_charttype[\_filters|N].png`
- Sidecar JSON: include filters, metrics, fallback notes, unclipped ranges

5) Visual Style & Accessibility

- Font: bundle open-source font (*Inter* or *Lato*)
- Palette: default to Viridis or ColorBrewer Set2
- Axis labels ≤ 12 pt, compact ticks
- Grid: light y-grid
- Legends: top-right
- Annotations: only notable points

6) Error Handling & Guardrails

- Fallback logic explicit in caption + JSON
- <3 rows: "Not enough data" card
- Clipping: P1–P99, but original min/max in JSON
- EV_disc_base preferred; fallback flagged

7) Caption & Alt-Text Autogen

- Caption: include scenario + filters, mean/median/total Δ EV
- Alt-text: one-sentence takeaway
- Footnote: "See Technical Note & README for methods."

8) Typical Calls

- Top-10 uplift (MPO)
- Histogram (Combined)
- Crossers by Sector (MPO)
- Side-by-side dumbbell (Combined)
- Scatter Δ p vs Cost (Override 0.5)

9) MVP Checklist

- Implement three first: Top-N bar, Histogram, Side-by-side dumbbell
- Enforce 1100 px width + compression
- Emit JSON with all metrics + metadata